Memo

To: [Employee]
From: Anthony Jordan
Date: September 22, 2025
Subject: Cantilever Arm Design with Tip Mass

Dear [Employee],

Thank you for your work on the last project! I'm glad to see that you are getting used to working at Burdell, Inc. There is never a dull moment here and we already have another project for you to work on, this time involving the design of a robotic arm with a payload under oscillating forcing. See the project brief for more details.

Project Overview:

Our client is bringing a compact robotic arm to production. The end effector mounts on a lightweight cantilever link that experiences a dominant periodic torque ripple from a gearbox at $\mathbf{60\ Hz}$. During recent pilot runs the arm exhibited audible vibration near that frequency. The client has asked us to quantify the arm's first bending natural frequency with a realistic payload attached and recommend a **minimum stock thickness** that reliably avoids resonance while controlling weight and cost.

We will model the link as an **Euler–Bernoulli cantilever beam** with a **lumped tip mass** M (tooling + fasteners). We are assuming small end deflection, uniform rectangular beam cross-section, no axial preload, and no damping in the beam. With the parameters beam length (L), beam width (b), beam thickness (t), Young's modulus (E), and density (ρ), the first bending frequency results from the **implicit characteristic equation** below:

$$1 + \cosh\lambda\cos\lambda = -\mu\lambda(\sinh\lambda\cos\lambda - \cosh\lambda\sin\lambda)$$

where

$$\lambda = \beta L, \quad \beta^4 = \frac{\rho A\omega^2}{EI}, \quad A = bt, \quad I = \frac{bt^3}{12}, \quad \mu = \frac{M}{\rho AL}$$

Once λ is found, the first natural frequency f_1 is found from:

$$\omega_1 = \beta^2\sqrt{\frac{EI}{\rho A}}, \quad f_1 = \frac{\omega_1}{2\pi}, \quad \beta = \frac{\lambda}{L}$$

Because this is a complicated set of equations, one of which is implicit, our senior analysts are suggesting we use **root-finding methods** to determine the natural frequency and resulting beam thickness recommendation. They have suggested **Bisection** (robust), **Newton–Raphson** (fast with a good slope), and MATLAB's `fzero` (robust, derivative-free). Please note that the above characteristic equation is likely to have multiple roots, which represent resonance frequencies at

different mode shapes for the bar. We are solving for the first, or fundamental, bending mode corresponding to the lowest valued non-zero, real root.

Your assignment is to:

(1) Determine the minimum thickness of the beam to meet the client specifications to reduce their manufacturing costs as much as possible. Recommend a viable stock from McMaster-Carr that would meet client needs based on your analysis.
(2) Perform a suite of convergence and performance analyses on our algorithms so the client may understand our methodology should they choose to perform this analysis given specification changes at a later time. We've also been asked to comment on any possible problems to avoid when performing our root finding analyses.

Client Specifications:

- Resonance avoidance: $f_1 \geq 1.25 \times 60\ \mathrm{Hz} = 75\ \mathrm{Hz}$ at nominal geometry and payload
- Geometry (nominal):
  - $L = 0.50\ \mathrm{m}$
  - $b = 30\ \mathrm{mm}$
  - $t = \text{decision variable}$
- Client's manufacturing division is requesting minimum viable thickness to keep costs in check.
- Material: Aluminum 6061 ($E = 68.9\ \mathrm{GPa}, \rho = 2700\ \mathrm{kg/m^3}$)
- Payload: Lumped tip mass (tool + adapter) $M = 1.20\ \mathrm{kg}$

Tasks:

1. **Determine beam thickness and source appropriate material**
   - Because our senior analysts suggest using a root-finding approach, you must recast the characteristic equation as a residual and find any supporting equations necessary to solve the root finding problem.
   - Please converge your beam thickness analysis to a relative percent specified tolerance of ε_s according to 5 digits of significant figure agreement in your algorithms. When using bisection, limit your search to values between $\lambda = [0.1\ 3.0]$. When using Newton-Raphson and `fzero`, use λ equal to the mean of the bracket values as your initial guess.
   - Our senior analysts suggest searching a range of bar thickness values between 20 and 80 mm.
   - Once the minimum appropriate beam thickness has been determined, report this value and suggest a viable selection of aluminum 6061 stock from McMaster-Carr for the client to use in their manufacturing process.
2. **Convergence Analysis**
   - The client would like to know the **iteration count** and **runtime** of each of the methods at our disposal for solving this problem as a function of specified percent

tolerance (a proxy for accuracy). They would like an analysis of iteration counts and runtimes for the bisection method, Newton-Raphson method, and the built-in MATLAB function **`fzero`** at the thickness determined in the previous analysis. Use the functions **`tic`** and **`toc`** for timing your algorithms.

- Please use the following parameters when performing your analyses:

Parameter	Value
$\varepsilon_{s,lower}$	10 significant figures of agreement
$\varepsilon_{s,upper}$	2 significant figures of agreement
Resolution of study	1000 points

- The client would also like you to predict possible problems with the algorithms should they choose to later perform the analysis themselves. **Please indicate at least one way that each algorithm may break** by finding a specific example for each (consider the impacts of changing initial value or interval bounds of λ).

Deliverables:

- A detailed PowerPoint presentation addressing the tasks above that adheres to client requirements and the Burdell Inc. style guide. See the provided template for reference. The slides must at least include:
  - A figure showing resonance frequencies vs. bar thickness.
  - A figure showing number of iterations of each method as a function of specified percent tolerance.
  - A figure showing runtime of each method as a function of specified percent tolerance
  - Supporting figures that show how changing the analysis bounds may affect the computational solutions with explanations of the observations.
  - Other figures are acceptable if they add clarity, actionability, or trustworthiness to the underlying analysis. Extra figures that do NOT sufficiently enhance these aspects of the presentation will confuse the client and be frowned upon.
- A .zip file containing your MATLAB scripts and supporting functions that produces all figures and calculations contained in your slides and adhering to the coding guidelines at Burdell Inc. Please make sure that your .zip file contains the following code elements:
  - A script file named **`beamAnalysis_[last name].m`** that calls all necessary subfunctions and produces all plots used in your presentation document, as well as produces all your numerical results.
  - A function **`[ e_s ] = significantDigitsPercentTolerance( n )`** which takes as input the number of significant digits, n, you would like for your relative percent specified tolerance and returns the percent specified tolerance, ε_s, that is

required for the desired level of accuracy. This will be the same function you have already written before.

- A function **`[ x_r, iter, time ] = bisectionMethod( f, x_l, x_u, e_s )`** which has the function handle, f, the lower bound, x_l, the upper bound, x_u, and percent error tolerance, ε_s, as inputs and the approximation of the root, x_r, the number of iterations, `iter`, and the runtime of the function, `time`, as the output.

- A function **`[ x_r, iter, time ] = newtonRaphsonMethod( f, dfdx, x_0, e_s )`** which has the initial guess, x_0, the function handle for the function, f, the function handle for the derivative, $dfdx$, and percent error tolerance, ε_s, as inputs and the approximation of the root, x_r, the number of iterations, `iter`, and the runtime of the function, `time`, as the output.

- Please be sure to check case and naming for all functions – they will be used in a larger coding environment with many contributors and any deviations from the above formatting will lead to errors. This will significantly impact my evaluation of your performance. The script must run in a fresh environment/directory with no variables pre-loaded into the workspace and must return the requested values and figures.

Please ensure that your deliverables are clear and concise, as I will be presenting them to the client. Because I haven't performed this analysis myself, I need you to provide detailed presenter notes that explain all your modeling choices, including but not limited to the context under which you are using your numerical frameworks, any relevant information for the client about how they work, and how you analyze the appropriateness and trustworthiness of the results. If you have any questions or need further clarification, do not hesitate to reach out to me.

I look forward to your insightful analysis and contributions to this project.

Best regards,

Anthony Jordan
Engineering Manager
Burdell Inc.